Welcome to

BIOL 5800V (CEMB 5900V) Genomic Data Analysis and Visualization

Instructor: Jeff Lewis
TA: Sonali Lenaduwe

Our goals as educators for this course...

1) Focus on the biology, while introducing students to fundamental concepts in genomics

2) Provide students with the framework to design carefully-controlled genomic experiments

3) Introduce the technology used to generate genomic data, and the strengths and weaknesses of current approaches

4) Introduce commonly-used methods in computational biology and bioinformatics

5) Provide hands-on experience in the basics of analyzing genomic data

6) Provide students with the intellectual framework and tools to delve deeper into analyzing their own data in the future

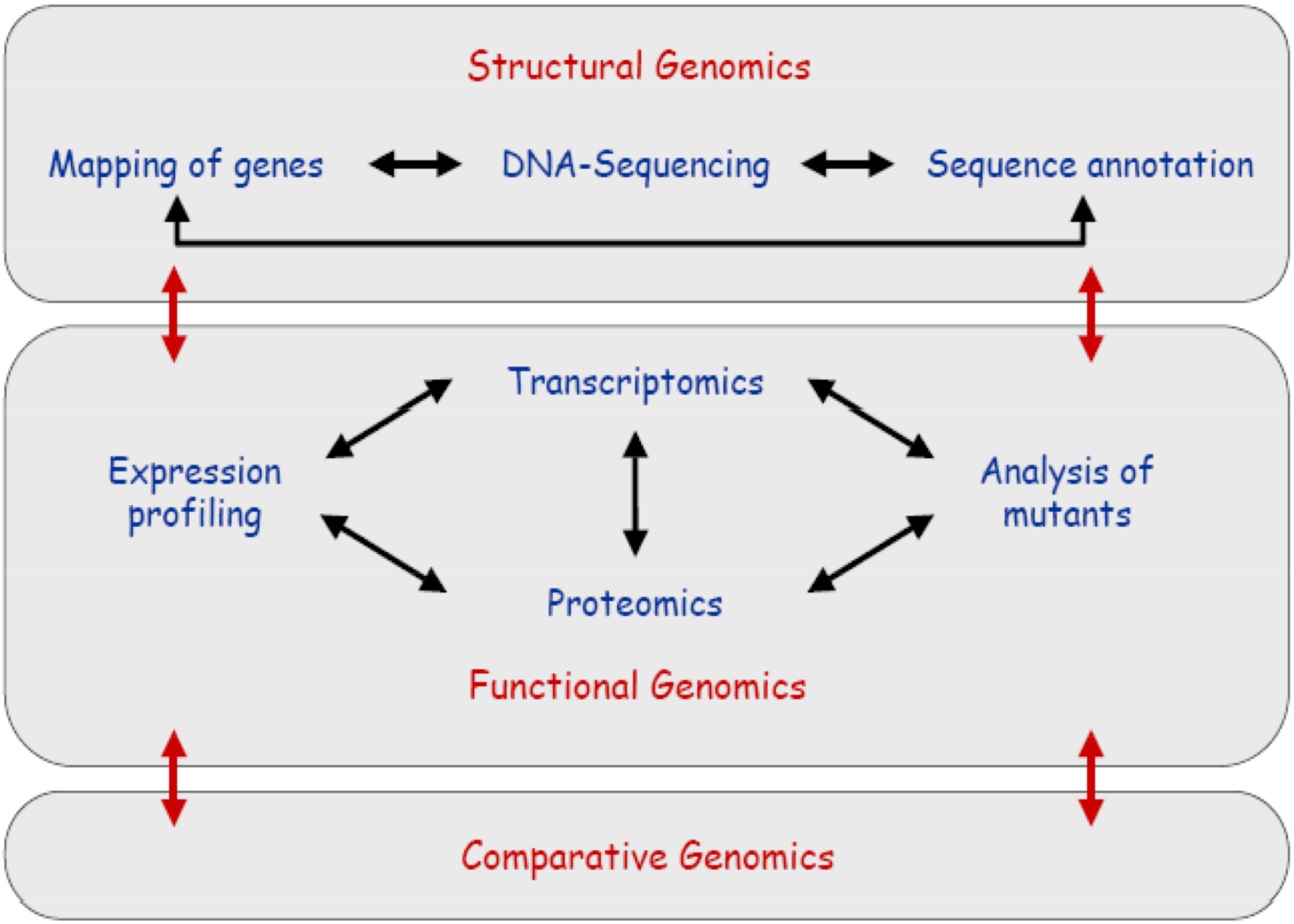
Defining Genomics

From the National Institutes of Health (NIH):

Genomics is a field of biology focused on studying **all the DNA** of an organism — that is, its genome. Such work includes identifying and characterizing all the genes and functional elements in an organism's genome as well as how they interact.

If genetics is (broadly) the study of genes,
genomics is the study of genomes

Three levels of genome research

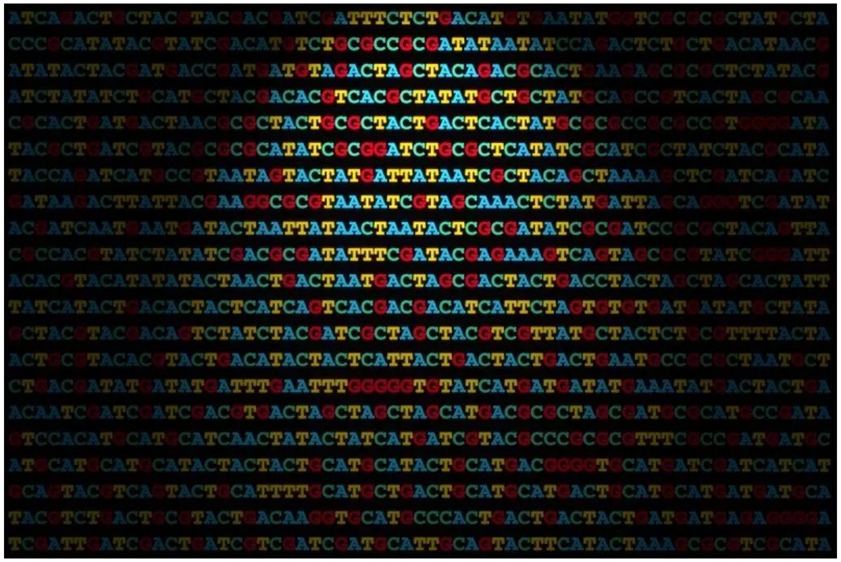


Types of questions that can be answered with comparative genomics

- How different are genome sequences between organisms or between individuals within the same species (phylogenetics)?
- What regions of the genome are most conserved across organisms, and what regions appear to be the newest?
- What regions of the genome appear to be under strong or weak selection?
- How do new mutations arise, and how do new mutations and standing genetic variation contribute to evolution?
- What is the genetic basis of complex traits (genetic mapping or genome-wise association studies (GWAS))?

Technical foundations of genomics

- Recombinant-DNA technology
- Molecular cloning
- PCR amplification
- Hybridization techniques
- DNA sequencing



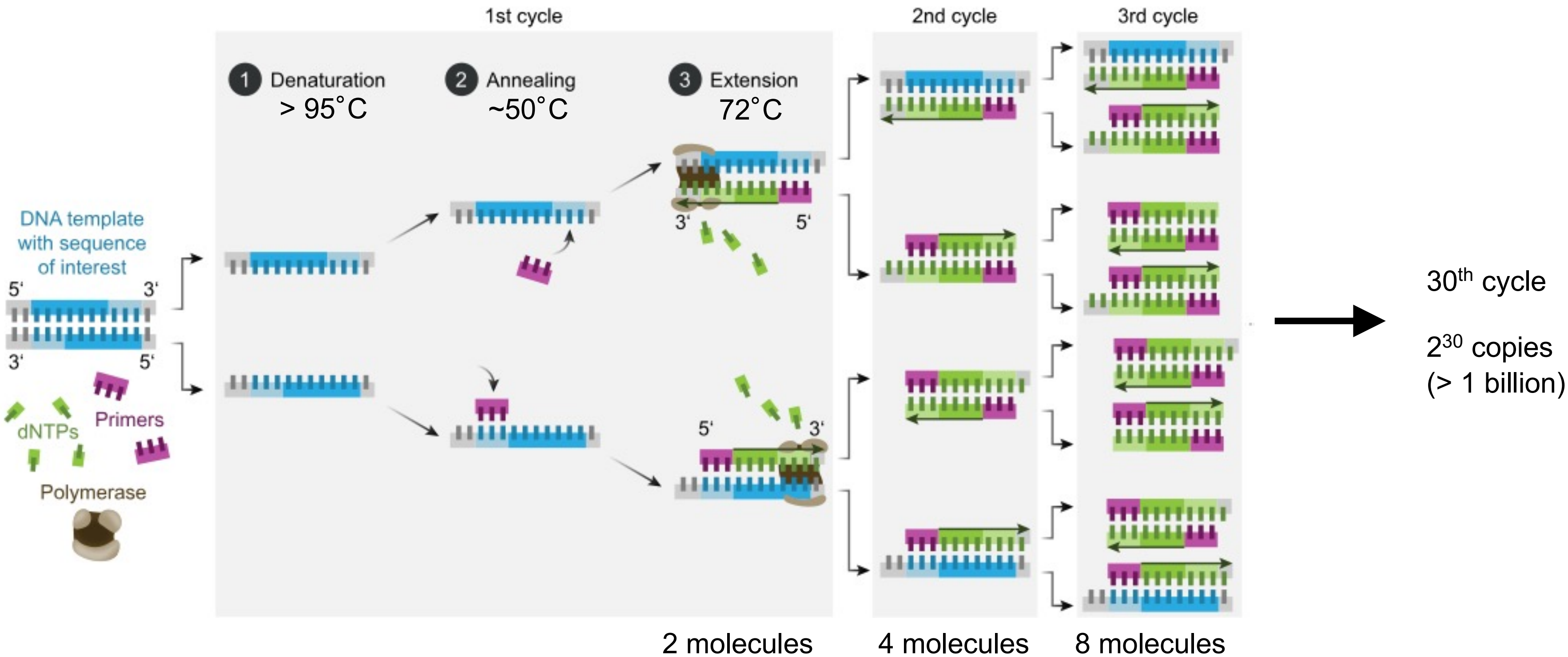
Memorial Sloan Kettering Cancer Center

Polymerase Chain Reaction (PCR)

- components necessary to amplify DNA for PCR:
  - buffer (pH optimized for Taq)
  - dNTPs (nucleotides used to synthesize DNA)
  - template (DNA to be amplified)
  - primers (small, synthesized DNA oligomers, specific to the template)
  - Taq (or another thermostable) polymerase

- automated “thermal cycling”
  1) denaturation (high heat renders DNA single stranded)
  2) annealing (primers bind to the template at lower temp.)
  3) extension (Taq extends DNA starting at the primers)

Polymerase Chain Reaction (PCR)



Wikipedia

DNA sequencing

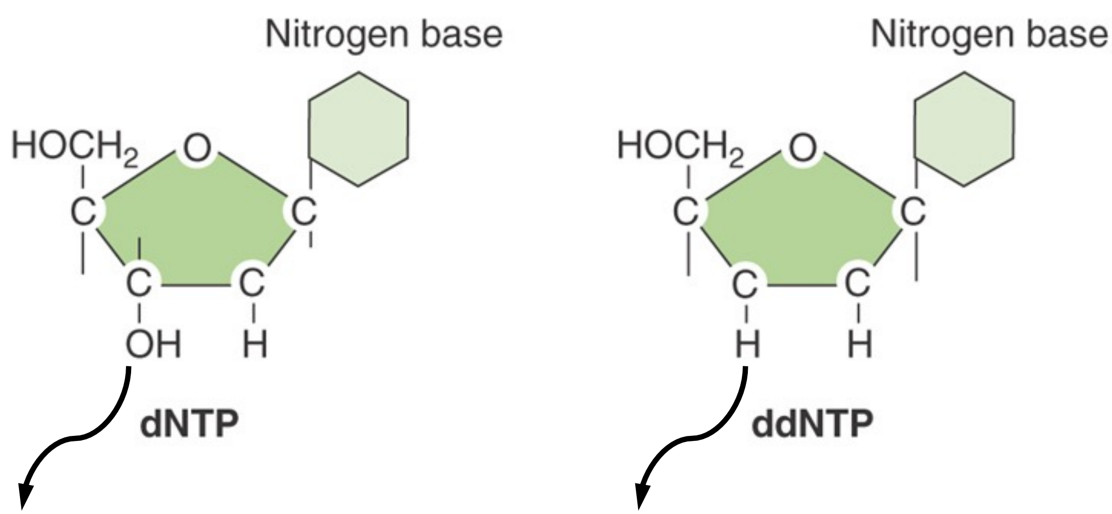
first generation: Sanger sequencing

second (next) generation: massively parallel (or high throughput) sequencing

third generation: long-read sequencing

Sanger sequencing: chain termination method

basic principle: DNA extension is stopped by dideoxynucleotides (ddNTPs)



attacks
triphosphate

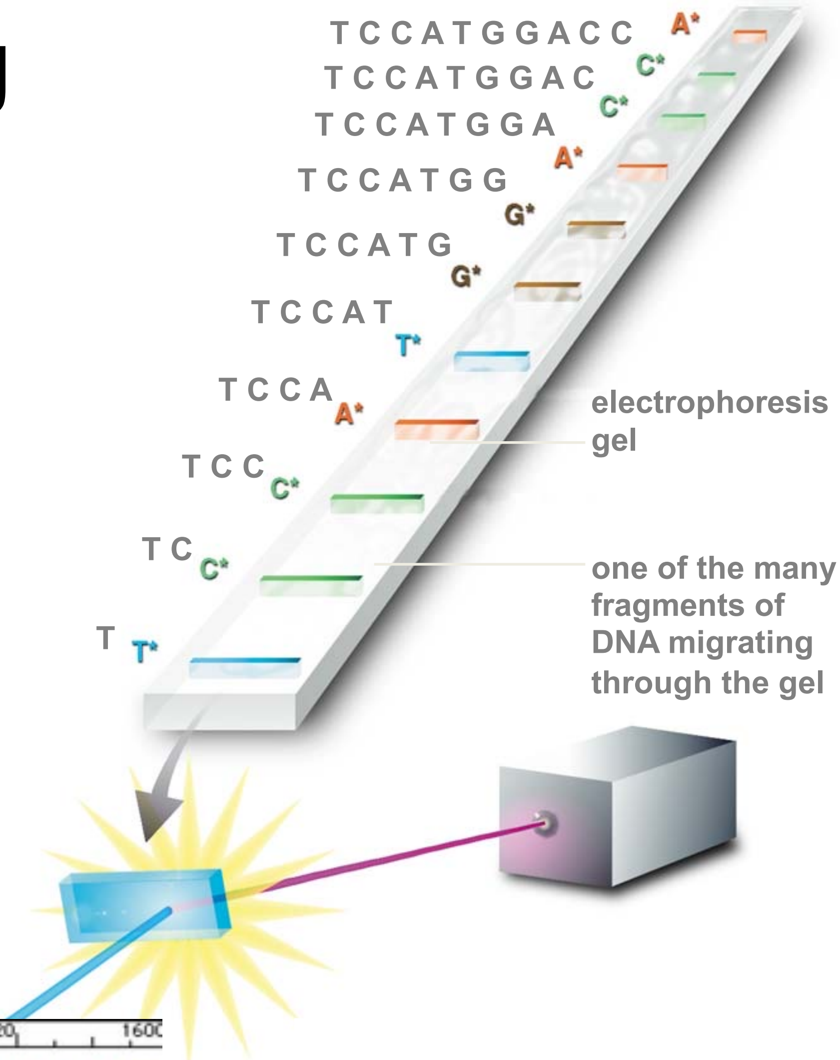
attacks
jack shittio

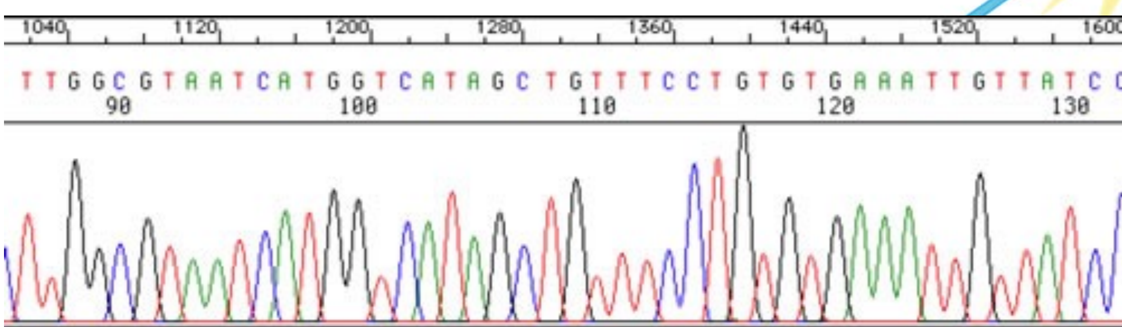
Sanger sequencing: chain termination method

- The 3′-OH group is missing in ddNTPs—this group is necessary for phosphodiester bond formation
- With addition of enzyme (DNA polymerase), a primer will be extended until a ddNTP is encountered
  - basically PCR with a single primer, so linear instead of exponential amplification
- The chain will end upon incorporation of a ddNTP
- With the proper dNTP:ddNTP ratio, you can get termination to generate fragments from 1-nt in length up to 750-nt
- conjugating the ddNTPs to different fluorescent dyes can allow for automated detection (e.g., A = red, T = blue, C = green, G = yellow)

Automated Sequencing

- DNA is placed on gel or capillary
- Fragments move off gel in size order; pass through laser beam
- Color each fragment fluoresces is recorded on printout





Second generation sequencing: sequencing by synthesis and reversible dye terminators

- Developed by Solexa (now part of Illumina)

1) A library is generated of fragmented DNA to be sequenced that have ”adapter” oligonucleotides attached to them

2) The library is loaded onto a flow cell containing a lawn of primers with complementarity to the adapters

3) DNA molecules that stick to a primer fold over to form a bridge, which is then PCR amplified using primers that are complementary to the adapter sequence

4) The process is repeated to generate amplified “clusters” of DNA across the flow cell (each cluster is unique)

Second generation sequencing: sequencing by synthesis and reversible dye terminators

5) clusters are sequenced using reversible fluorescent terminators, so DNA polymerase adds a single base to each cluster at a time

6) An image is generated of the flow cell (one for each color) to denote the first nucleotide sequence

7) The reversible fluorescent terminator is cleaved off and washed away, and the second reversible terminator is added

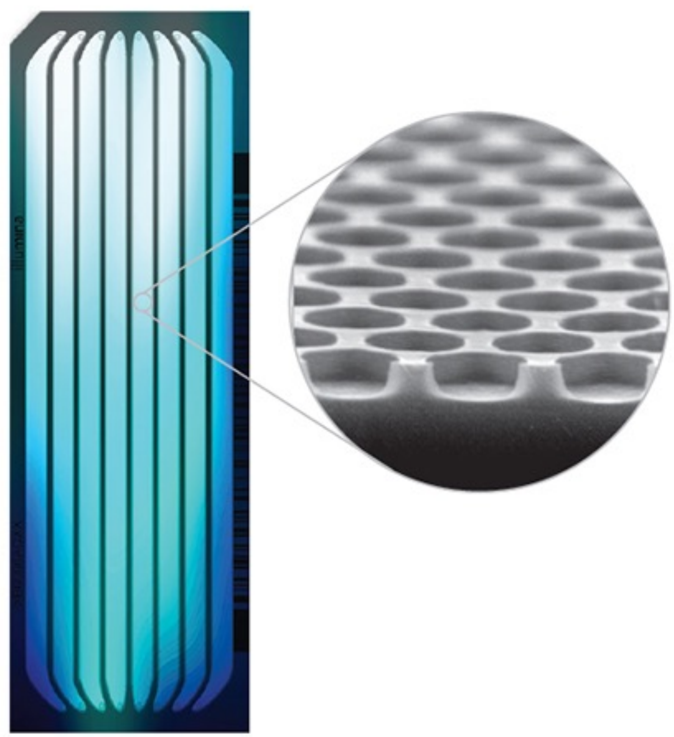
8) An image is generated for the second nucleotide, and the process continues (for 50 to 150 cycles, leading to 50 – 150 bp of sequence)

9) The reverse strand can be sequenced as well, which is called "paired-end" sequencing (e.g., 50 x 2 bp paired-end sequencing)

Illumina Sequencing

- current technology using “patterned” flow cells, where essentially a single DNA molecule forms a cluster and is sequenced in each well

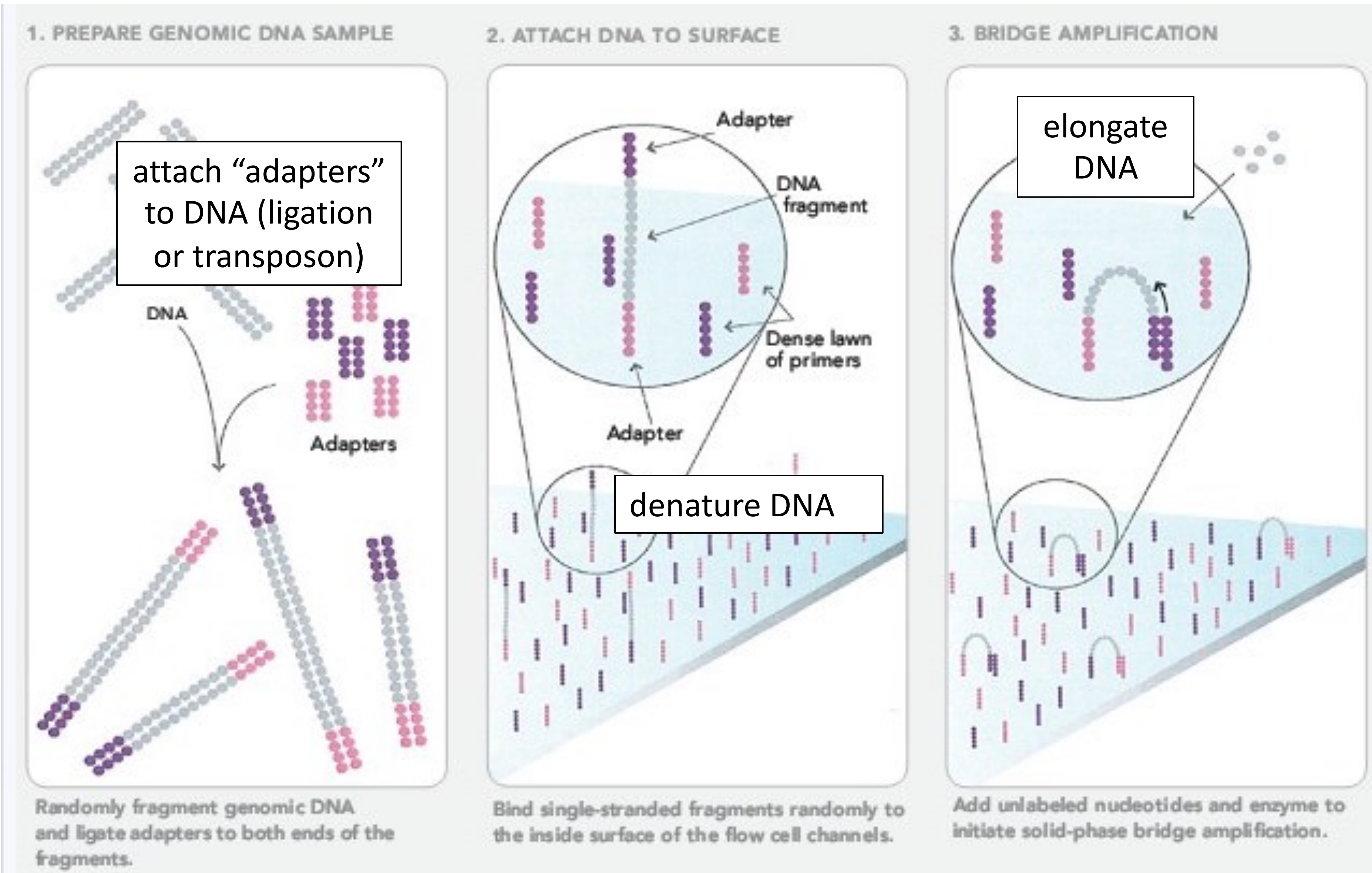
flow cell with
8 lanes
(allowing for
8 independent
samples)



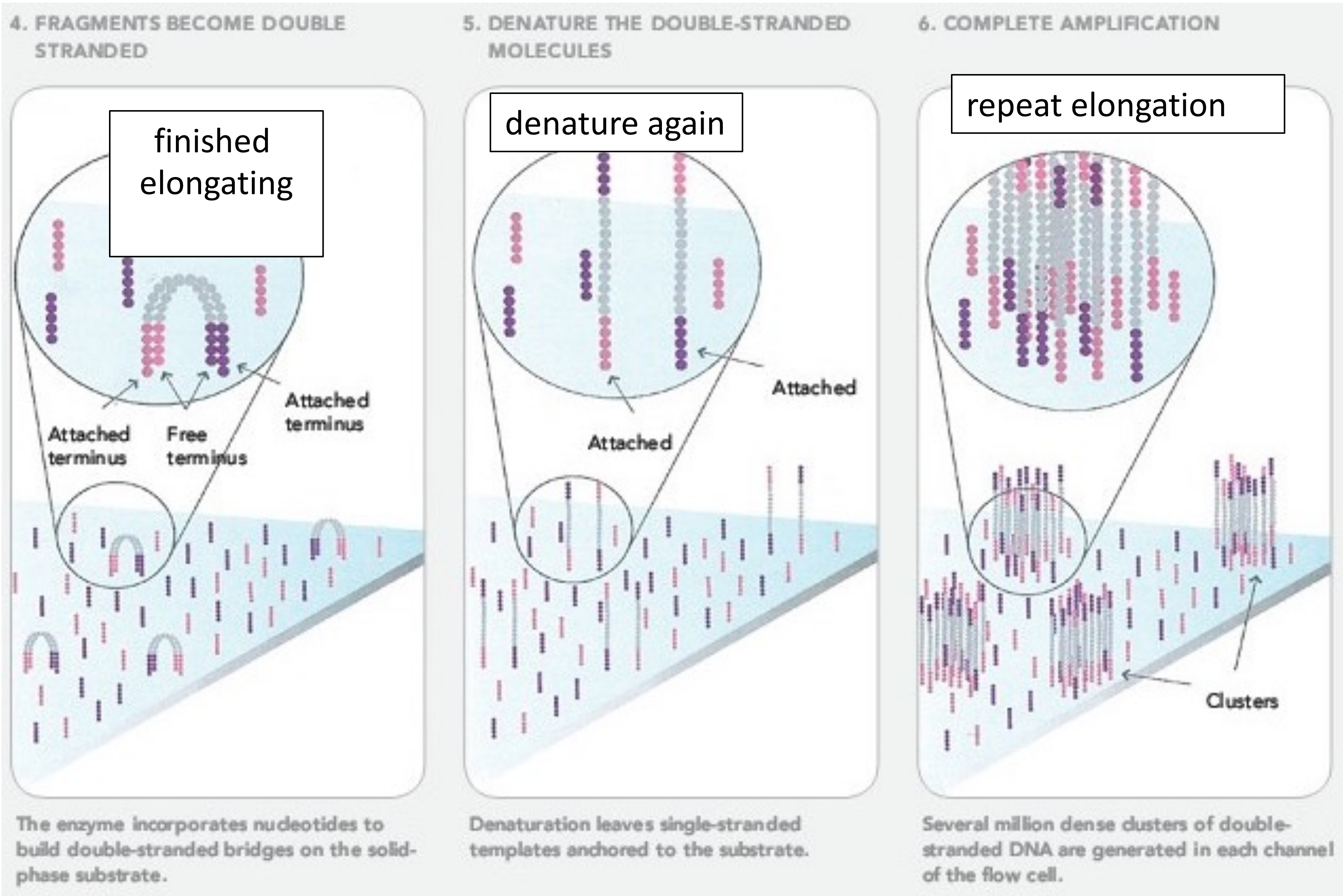
nanowells that
ensure even density

Illumina

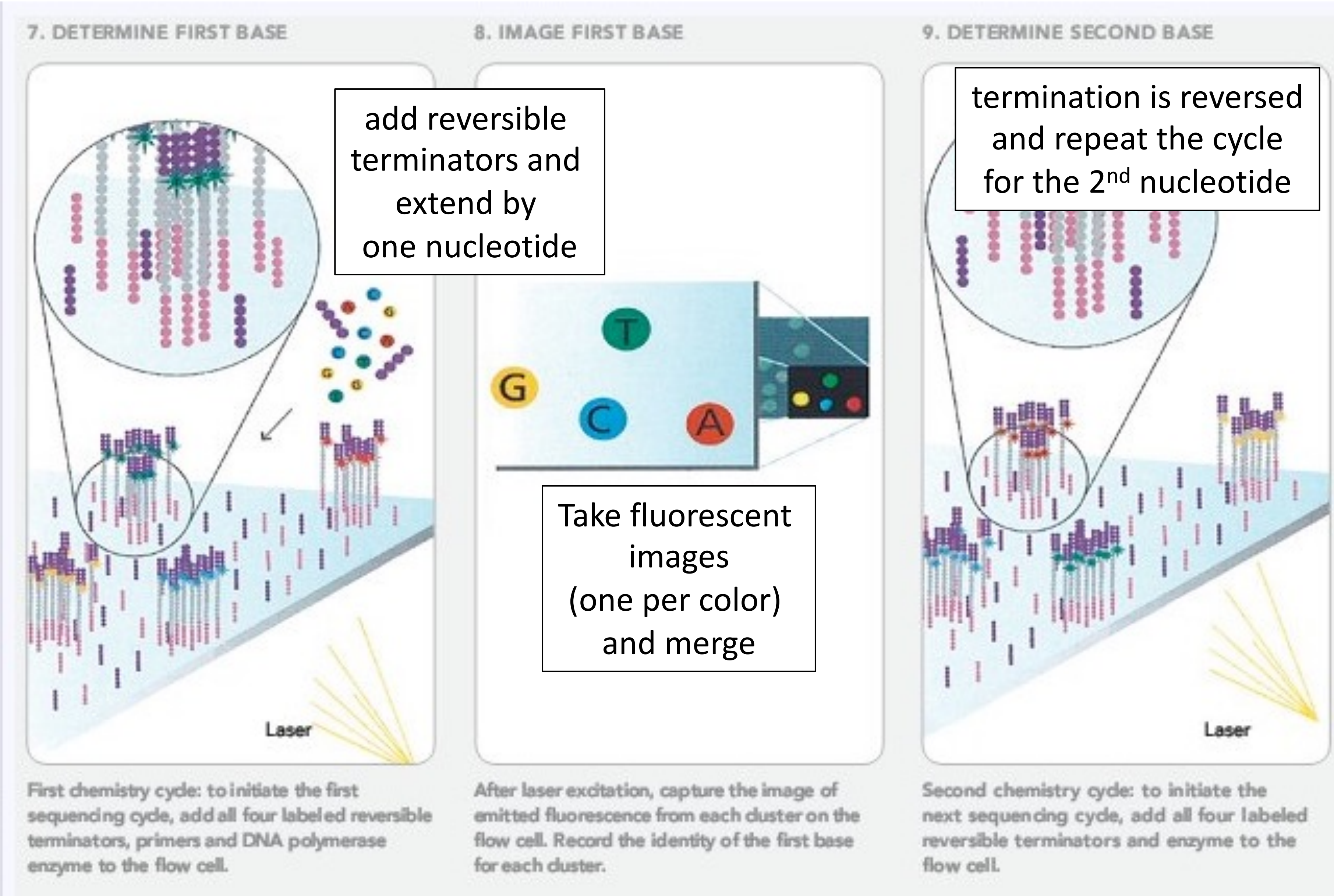
Illumina Sequencing



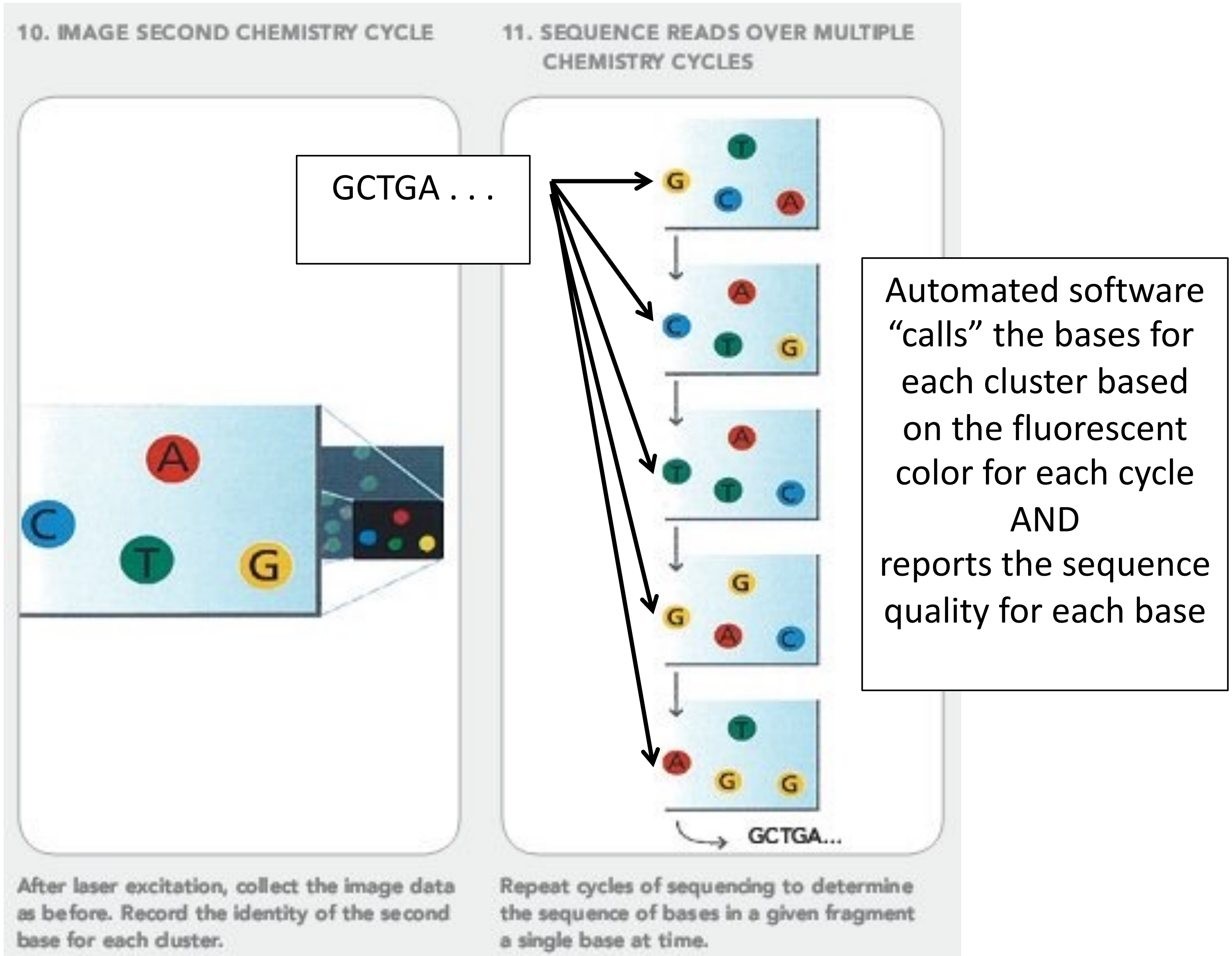
Illumina Sequencing



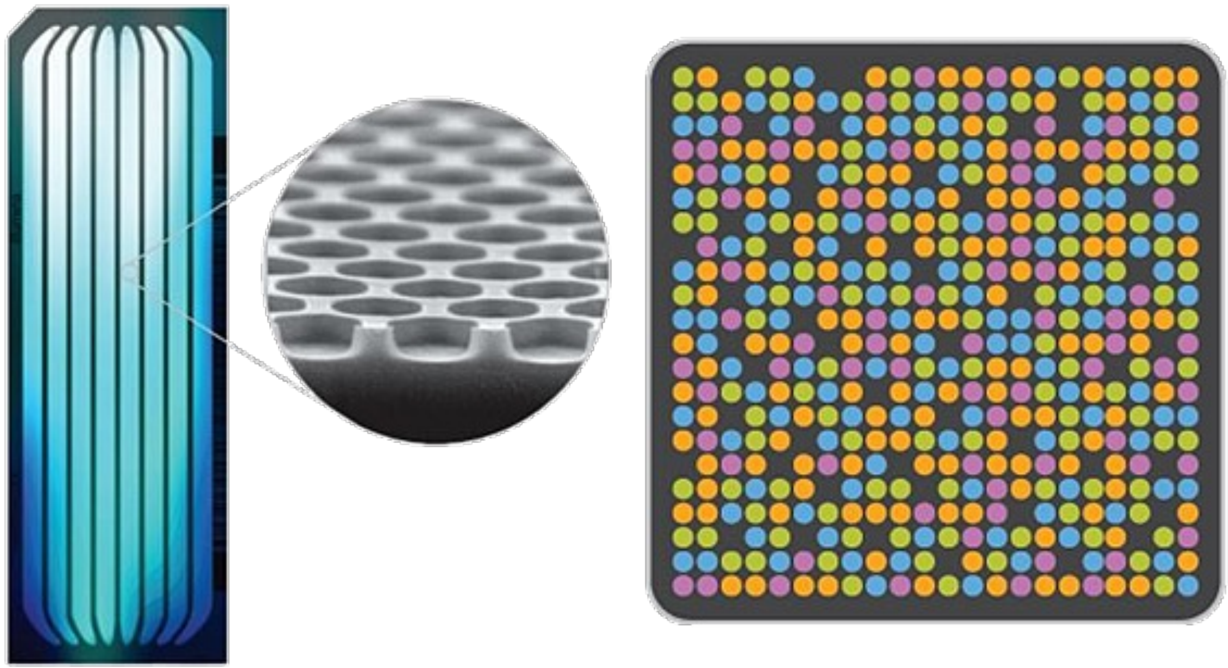
Illumina Sequencing



Illumina Sequencing



Illumina Sequencing



Final output: millions of short (50-150bp) single end (sequenced just the 5' part of each DNA fragment) or paired-end (sequenced both ends) reads

current technology (NovaSeq X) is up to ~25 billion clusters per flowcell (enough to sequence >60 human genomes) in ~48 hours

Third generation: long read sequencing

- Two main types: PacBio sequencing and Oxford Nanopore sequencing
- Generate hundreds to thousands of long (many kilobases or kb) reads
- Generally, much less accurate than short-read sequencing (high end of 99% accuracy vs. 99.9% or higher for Illumina)
- But, great for “end-to-end” chromosome sequencing

PacBio Sequencing

1) DNA is circularized by ligating adapters to each strand

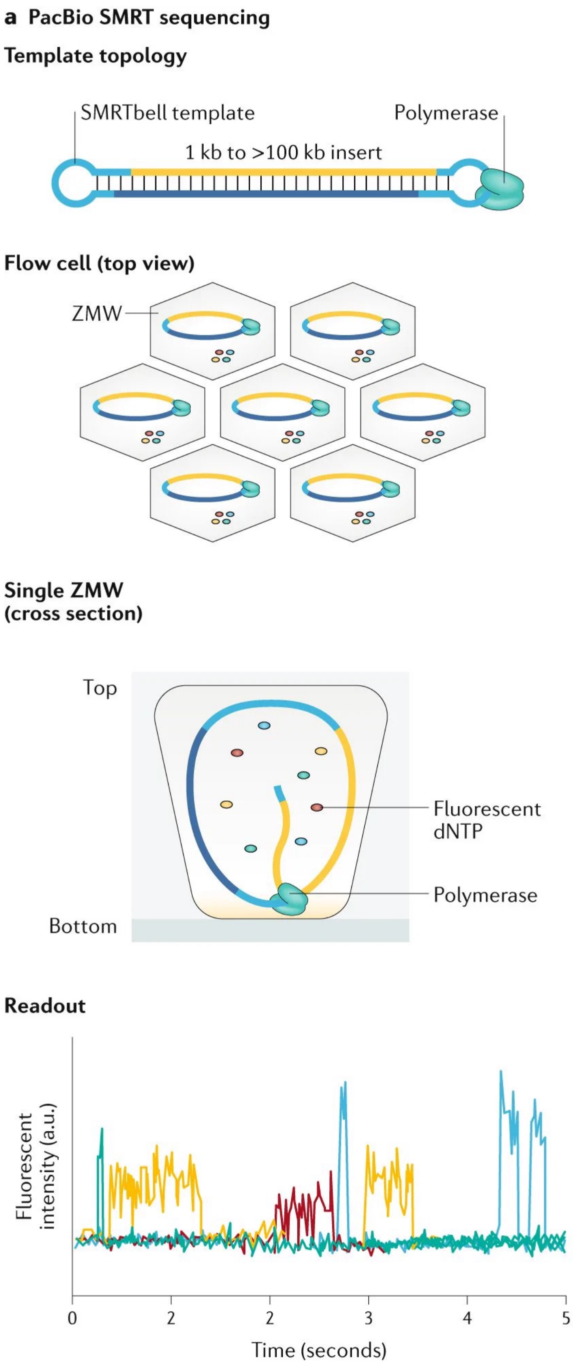
2) DNA polymerase and a primer (homologous to the adapter) are added to the DNA

3) The circularized DNA molecules (plus Pol and primer) are added to a flow cell and are immobilized in a nanowell

4) Fluorescent nucleotides are added, and each nucleotide incorporated emits light and is measured for each well

5) The DNA sequence is read repeatedly (since it’s circular) to increase accuracy, and reads average 10-25 kb

PacBio Sequencing



Logson et al. 2020. Nature Reviews Genetics. 21:597-614

Oxford Nanopore sequencing

1) DNA (or RNA) is directly sequenced after passing through a nanopore channel

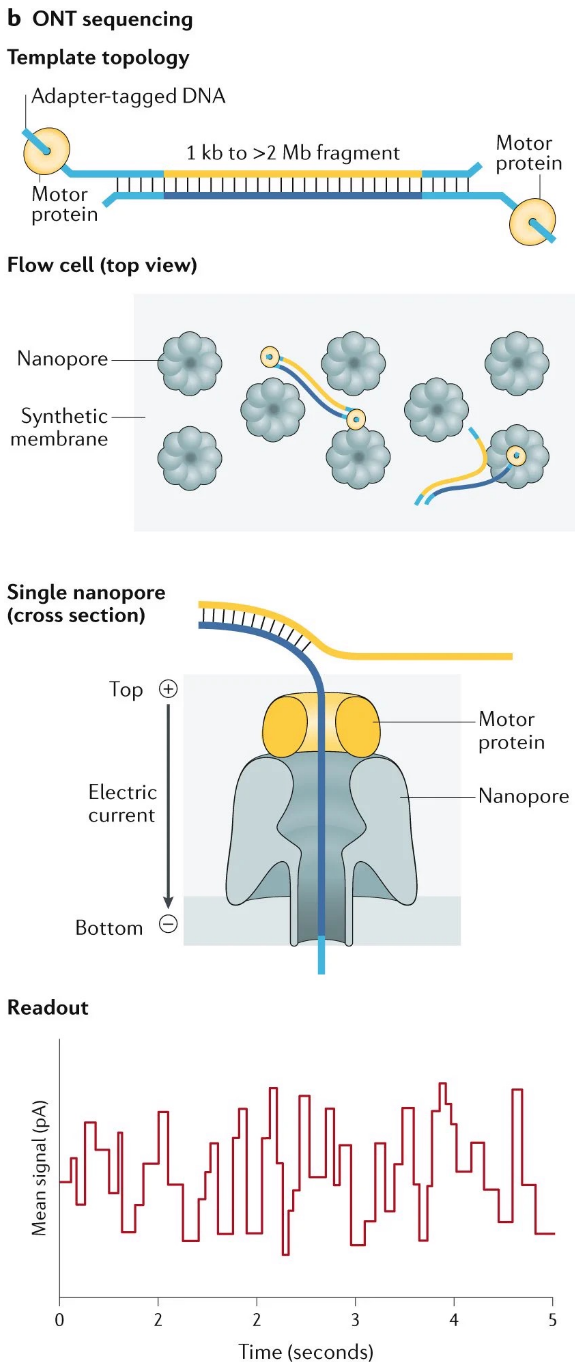
2) Protein nanopores are embedded in a synthetic membrane

3) A helicase unwinds the DNA and and feeds one strand through the pore

4) A current is applied, with each nucleotide base causing a characteristic disruption of the electrical current

5) The readout is 50,000 to 100,000 unique reads with an average length of 30-kb but up to 2-mb (million bases)

Oxford Nanopore sequencing



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This course will focus on short-read (highly-parallel) sequencing

A “standard” Illumina sequencing workflow

1) obtain nucleic acids (DNA or RNA)

2) generate Illumina library
    - fragment nucleic acids (~300-bp)
    - ligate adapters (or transposon “tagmentation,” which both fragments and ligates adapters at the same time)
        - Note: can use “barcoded” adapters where each sample receives a unique known DNA barcode for multiplexing
    - PCR amplify library

3) sequence on flowcell (performed at dedicated sequencing centers)

4) receive millions or reads and their quality scores per nucleotide for each sample

A “standard” Illumina sequencing workflow

5) perform quality control analysis of each sample

6) align reads to a known reference genome (or transcriptome), or assemble the genome de novo (if it’s completely new)

7) assign reads to genes or generate annotations de novo to identify genes in your new genome

8) perform downstream analysis based on why you sequenced the samples
    - e.g., for RNA-seq, determine the number of read counts that map to each gene and see how those differ between samples
    - e.g., for DNA, identify allelic differences between samples to identify polymorphisms

Challenges in genomic analysis

- how to handle enormous amounts of data (everything from storage to computational time to organization)
- how to develop robust computational algorithms that accurately capture biology
- reproducibility at both the bench and at the computer
- avoiding “batch” effects (technical variation that occurs due to differences in sample handling across “batches”)
    - e.g., libraries generated by two different researchers, RNA isolated by different methods, sequencing on different lanes or flowcells

F1000Research
F1000Research 2015, 4:121 Last updated: 28 MAR 2022

Check for updates

RESEARCH ARTICLE

A reanalysis of mouse ENCODE comparative gene expression data [version 1; peer review: 3 approved, 1 approved with reservations]

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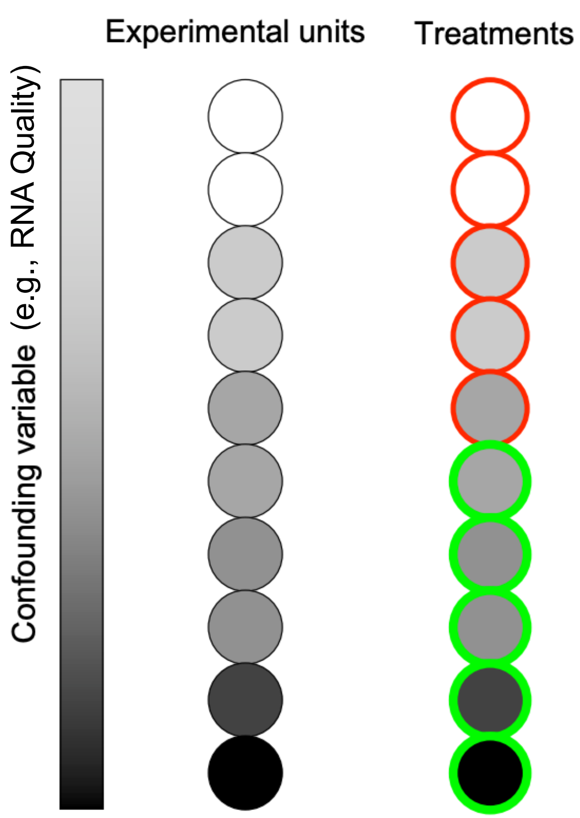
Abstract

Recently, the Mouse ENCODE Consortium reported that comparative gene expression data from human and mouse tend to cluster more by species rather than by tissue. This observation was surprising, as it contradicted much of the comparative gene regulatory data collected previously, as well as the common notion that major developmental pathways are highly conserved across a wide range of species, in particular across mammals. Here we show that the Mouse ENCODE gene expression data were collected using a flawed study design, which confounded sequencing batch (namely, the assignment of samples to sequencing flowcells and lanes) with species. When we account for the batch effect, the corrected comparative gene expression data from human and mouse tend to cluster by tissue, not by species.

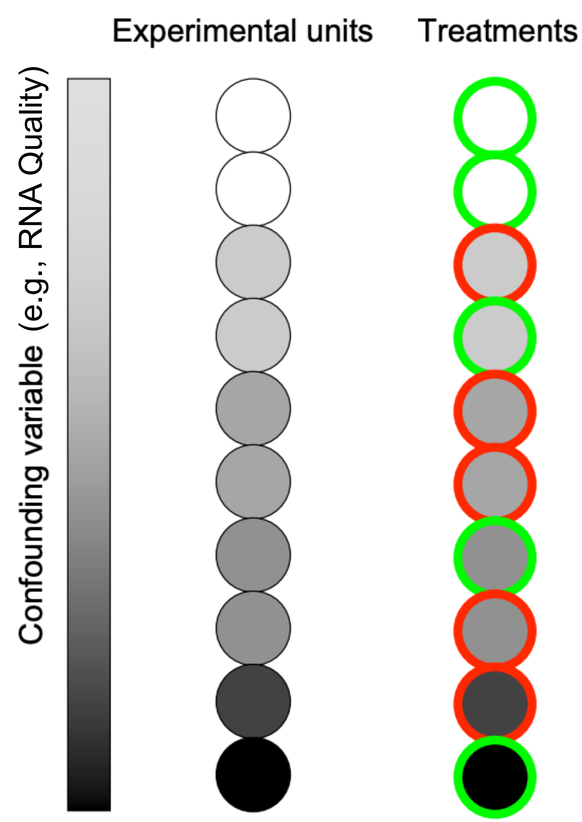
Batch effects and other confounders

- If the confounder completely correlates with your experimental treatment, you will not be able to correct for it

Non-randomized design: confounder is variable across treatments



Randomized design: confounder is evenly represented between treatments



Best upfront practices

- determine how many samples you need for a comparison (e.g., a minimum of 3 biological replicates per experimental group for RNA-seq if using genetically identical individuals)
- perform a power analysis if unsure of the necessary population size (do you want to be able to detect genes with at least a 2-fold difference in expression?)
- capture day-to-day variation (e.g., collect all of Replicate 1 samples on Day1, all of Replicate 2 samples on a separate Day 2, etc.)
- randomize how samples are processed and have a balanced experimental design (e.g., make sure that your treatments and controls are not all separately or sequenced on different lanes)

For Thursday and Next Tue...

- We'll have our first lab exercise, and introduction to the command line and R
- Next Tuesday, Andy Alverson will give a guest lecture on the Arkansas High Performance Computing Cluster (AHPCC).
- You will receive an email today with a link and instructions for how to get an account for AHPCC.